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Ch=05 Software Tools

* Software tool ? A software tool ~~to~~ used to create, maintain, or otherwise support programs & applications.

1) Specification :- specify the GUI

2) Grammer :- written instruction program would understand.

3) Tranpaolion diagram :- Set of nodes & linkes displayed in text or state diagram.

4) Statecharts

5) Interface building tool :- Design tool that helps to design ⇒ data entry structure, widgets

6) Interface mockup tool :- Tool to develop a quick sketch of GUI ⇒ Visual studio, Pen, paper

7) Software engineering tool :- Provide info. about software that ~~to~~ developed.

8) Evaluation tool :- Tools to evaluate the correctness & completness of programs.

* Specification method :- Default language used for specification is natural lang : English.

→ Other lang like formal & semi-formal lang used bcz natural lang is lengthy.

→ Various specification method:

1) Grammers :- It is useful to specify ~~textual~~ expressions that a program should understand.

2) Menu-selection & dialog-box tree

Specification methods include online tools to help construction of menu tree.

3) Transition Diagram :- Transition diagram has a set of nodes that represent system states & a set of link b/w nodes

4) Statecharts : Same part of Diagram.

* Interface building tool :- Allow Architects, Designer, Managers, user, & engineer to sit and Discuss the design.

→ Paper based designs are best

→ Various software tool :-

a) Interface Mockup tool :- User interface mockups can be created using pencil, word processors, slide show presentation.

b) Software engineering tool :- choosing lang Java, C, C++, C# sometime a complex task.

c) The windowing system layer: some platform are new it can't afford ~~hity~~ high level building tool. engineer have to work on low level - icons, image, fonts.

d) The GUI toolkit layer :- Offer flexibility to programmer

→ Difficult to use without interactive support

→ GUI toolkit offer widgets
- → Scroll bars
- → pulldown, popup menus.

* Interface building tools (feature).

1) User interface Independence :-
- → Enable multiple user interface
- → Enable multiple platform support
- → Separate Interface design from Internals

2) Methodology & notation :-
- → Develop design Procedures
- → find ways to talk about design.

3) Rapid Prototyping :-
- → Try out idea very early
- → Test revise, test, revise
- → Engage user, manager, customer.

4) Software support :
↳ Ease maintenance

* Interaction Devices :-

1) Touch screen :- By the finger or stylus.
→ Very fast, no specilised pointing device

2) Gesture recognition :- Understanding human movement
→ Computer understand our movement

3) Speech Recognition :- Process of converting ~~written~~ spoken phrase into written text

4) Keyboard : Through keys.

5) Response time :- Time taken by a device to respond to a request.

* Describe Keyboard :- Type writer (old)
QWERTY Keyboard (we use).

→ Function Keys :- Programmed to perform som task.

Ex
F1 : Display help screen
F2 : Highlight screen F5 : Refresh content
F3 : Open search tool

* Describe Pointing Device :- It is Useful for 6 type of interaction

1) Select :- User choose from set of item

2) Position :- To drag a block of text.

3) Orient :- Indicate a direction of motion.

4) Path :- A curving line in a Drawing Program.

5) Quantify :- User specify a numeric value.

6) Text :- User enter move or edit text.

* Type of Pointing Devices :-

1) Direct control pointing Device :-

a) Light pen :- To point a spot on screen

b) Touch screen

c) Tablet PCs & mobil device

d) Stylus

2) Indirect control Point :-

1) Mouse 2) Trackball.

3) Touch pad :- In laptop mouse (Touch screen)

4) Pointing stick, like a Joystick

* Speech Recognitions:-

1) Discrete-word recognition:- It recognize individual words spoken.

→ Quiet environment, head-mounted microphone improve recognition rate.

2) Continuous-speech recognition:- Allow the user to talk without a pause.

→ Ability to understand the astronauts spoken words.

→ Error rate & error repair are serious problm.

3) Voice information system:- These voice Info. systems often called Interactive Voice Response (IVR) provide good customer service at low cost.

Ex Jio call service.

4) Speech generation:- Artificial simulation of human speech with by a computer or other device.

5) Non-speech auditory interface, Include individual audio with more complex info. presentation

→ Indicate warning / acknowledge the completion

Ex Keyboard, Mouse.

* Image & video Display :-

→ Rapid operation → Reasonable size

→ Quiet operation → No paper waste

→ Reliability → Reasonable Resolution

→ Graphic & animation

2) For certain application, monochrome display are best bcz of low cost.

3) Display technology :-

1) CRT (cathode Ray tube) :- evacuated glass tube

→ A beam of electron pass & hit phosphor coated screen to generate picture

2) Plasma panel :- Mixture of gas include neon.

3) LCD (Liquid crystal Display) :- Passing Polarized light through liquid crystal.

4) Flat panel display :- Our display,

* Characteristics of video display Devices

1) Persistence :- Time it takes the emitted light from the screen to decay to 1/10th of its original intensity.

2) Resolution :- No. of points / centimeter.

3) Aspect Ration :- It is the ratio of vertical to horizontal point. Ex 16:9

* Driver :- Is a program that interacts with a particular device or software

→ It contain special knowledge or special software interface.

→ Examples In running a game / software.

Thank you

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